

Cover Letter for Carson Slater

To the hiring manager of the actuarial department of Texas Farm Bureau Insurance Company,

I am a fourth-year Ph.D. candidate in the statistical science department at Baylor University writing to apply to a part-time data analyst position at Texas Farm Bureau Insurance Company. After talking with Blake Brooks at the Baylor University career fair, I realize that Texas Farm Bureau Insurance's mission to help policyholders manage the financial risks of everyday life with high ethics and timely professionalism is a mission I can gladly stand behind. After speaking with Trey Schneider, it is evident to me that Texas Farm Bureau Insurance Company is a great place to work with a culture of professionalism and caring for its employees. As I finish my Ph.D., I am looking for an opportunity to apply the various skills and statistical methods I have learned in my coursework and consulting opportunities, and this analyst role in the actuarial department seems like the ideal environment to begin applying my expertise.

As a current graduate student, a part-time position is a perfect arrangement for me to use my skillset to contribute to Property and Casualty insurance projects while finishing my doctoral dissertation. I thrive in environments where I am working on multiple projects at the same time. In my time at Baylor, I have been able to work on contract projects with Denver Water to develop methods for understanding single-family residential customer water usage. It involved developing data processing pipelines, combining different types of data pertaining to customer usage and customer features, and crafting tailor-made solutions and dashboards to assess and inform Denver Water's residential conservation efforts. I was able to uncover that roughly 20% of the Denver Water customer base does not irrigate their outdoor spaces. This information led them to reconsider their conservation strategy to focus more on targeted messaging.

I have no experience in the insurance industry; however, I am someone who is able to teach myself both abstract theoretical topics and deeply technical skills. Whether it is learning to scrape weather data for modeling purposes, or learning proper prior specification on a covariance matrix for a Bayesian model in a specific actuarial context, I am able to get into the weeds on both sides of the coin and explain these concepts to technical collaborators and nontechnical stakeholders. A canonical example of this is when I consulted for the American Holistic Nursing Association, where I used a mixed-effects model for longitudinal survey data to assess patient improvement using different treatment frameworks. In that project I collaborated with graduate students as peers and present our model findings in a parsimonious fashion to the stakeholders. Despite my lack of insurance experience, my Ph.D. training has taught me to be able to teach myself nearly anything.

This part-time analyst role demands an individual who deeply understands both the nuances of statistical methodology and the skills required to leverage big data. It also requires self-motivation and time management, which are prerequisites for completing a Ph.D. My background has equipped me to translate complex modeling into objective, actionable insights that have the potential to make a meaningful impact on property and casualty insurance projects.

Thank you for your time and consideration. I would welcome the opportunity to discuss how my background makes me a great fit for this role.

Sincerely,

Carson Slater

Ph.D. Candidate, Statistical Science
Baylor University